

EXPRIMENT-3

3.) Write a C program for Playfair algorithm is based on the use of a 5 X 5 matrix of letters constructed using a keyword. Plaintext is encrypted two letters at a time using this matrix.

PROGRAM:

```
main.py Visualize Run ▶
1 def create_matrix(key):
2     key = key.upper().replace("J", "I")
3     alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
4
5     letters = ""
6
7     for ch in key + alphabet:
8         if ch.isalpha() and ch not in letters:
9             letters += ch
10
11     matrix = [letters[i:i+5] for i in range(0, 25, 5)]
12     return matrix
13
14
15 def find_position(matrix, ch):
16     for i in range(5):
17         for j in range(5):
18             if matrix[i][j] == ch:
19                 return i, j
20
21
22 def encrypt(text, matrix):
23     text = text.upper().replace("J", "I")
24     result = ""
25
26     i = 0
```

```
main.py Visualize Run ▶
26     i = 0
27     while i < len(text):
28         a = text[i]
29
30         if i + 1 < len(text):
31             b = text[i + 1]
32         else:
33             b = "X"
34
35         if a == b:
36             b = "X"
37             i += 1
38         else:
39             i += 2
40
41         r1, c1 = find_position(matrix, a)
42         r2, c2 = find_position(matrix, b)
43
44         if r1 == r2:
45             result += matrix[r1][(c1 + 1) % 5]
46             result += matrix[r2][(c2 + 1) % 5]
47
48         elif c1 == c2:
49             result += matrix[(r1 + 1) % 5][c1]
50             result += matrix[(r2 + 1) % 5][c2]
```

```
main.py Visualize Run ▶
45     result += matrix[r1][(c1 + 1) % 5]
46     result += matrix[r2][(c2 + 1) % 5]
47
48     elif c1 == c2:
49         result += matrix[(r1 + 1) % 5][c1]
50         result += matrix[(r2 + 1) % 5][c2]
51
52     else:
53         result += matrix[r1][c2]
54         result += matrix[r2][c1]
55
56     return result
57
58
59 key = input("Enter the key: ")
60 plaintext = input("Enter the plaintext: ")
61
62 matrix = create_matrix(key)
63
64 print("5 x 5 Matrix:")
65 for row in matrix:
66     print(" ".join(row))
67
68 ciphertext = encrypt(plaintext, matrix)
69
70 print("Ciphertext:", ciphertext)
```

OUTPUT:

```
Enter the plaintext: HELLO
Ciphertext: ITSSG
```